

Availability-Aware Multi-Signal Fusion for Detecting Machine-Generated Images and Video

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Abstract—Fully synthetic images and video are now produced by hundreds of publicly available generators, and a detector that relies on any single cue fails whenever that cue is absent. We present VidTrust AI, a detector that fuses three deliberately independent signals—a pretrained neural classifier, provenance metadata (EXIF/XMP/C2PA), and a fast Fourier transform (FFT) high-frequency energy ratio—under an *availability-aware* fusion rule: a signal that cannot run is removed from the weighted average and the remaining weights are renormalised, rather than being scored as zero. On a 400-image balanced subset of Community Forensics spanning 50 generators, the fused system reaches ROC-AUC 0.9196, against 0.8397 for the classifier alone and 0.9012 for the frequency signal alone. We further show that the standard evaluation protocol conflates synthesis artefacts with resampling history: under a centre-crop control that equalises processing history without interpolation, AUC rises to 0.9487 while recall is unchanged to four decimal places—the entire gain is a reduction in false positives on real photographs, from 70 to 39. We report two negative results that only per-signal logging exposed: a three-character provenance signature that fired at full confidence on 62 of 200 real photographs, and an earlier classifier under which the fused ensemble ranked *below* its lowest-weighted component. Finally, we measure an operating point that improves held-out accuracy from 0.8342 to 0.9664 and decline to adopt it, because it is fitted to a dataset confound we can name. We argue that per-signal score dumping and ablation should be treated as a minimum reporting standard for fused detectors.

Index Terms—synthetic media detection, AI-generated image detection, sensor fusion, content provenance, C2PA, ablation study, evaluation methodology

I. INTRODUCTION

The manipulated media that most needs checking is no longer a face swapped into an otherwise genuine photograph. It is media that is synthetic in its entirety: an image or clip that no camera ever recorded. Latent diffusion models [1] and their thousands of community fine-tunes have made such media cheap, and the detection problem has shifted with it. A face-swap detector [12] needs an original to compare against; a fully synthetic image has no original, and frequently no photographed face at all.

Detectors for this problem are commonly single-model classifiers [2]–[4]. That design has a structural weakness that is easy to miss in benchmark numbers: it has exactly one way to be wrong, and nothing to check it against. An ensemble of independent signals can in principle cover that gap—but only if it handles the case that motivates it. Provenance metadata

is the strongest available evidence of synthetic origin when it is present, and it is absent from the great majority of media in circulation, because every social platform re-encodes on upload and strips it. A fusion rule that scores a missing signal as 0.0 treats “no evidence” as “evidence of authenticity” and pushes every stripped file toward a verdict of real.

This paper makes four contributions.

- 1) **An availability-aware fusion rule** (Section III-B) under which an inoperative signal is excluded from the weighted average and the remaining weights renormalised, with total signal failure mapping to maximum ignorance rather than to either verdict.
- 2) **A two-track evaluation protocol** (Section IV) that refuses to average a large public set on which the provenance signal is structurally dead with a small hand-collected set on which it is the only thing being tested.
- 3) **A resolution-normalisation control** (Section IV-D) showing that a naive protocol on the standard public set measures resampling history in addition to synthesis, and quantifying how much of the reported performance that accounts for.
- 4) **Two reported negative results** (Section VI) that were invisible to unit tests and to aggregate metrics, and were surfaced only because the evaluation harness dumps every signal’s score and availability flag per file.

We state at the outset what this work does not claim. No model is trained or fine-tuned here; the classifier is used exactly as published. The fusion weights and verdict thresholds are hand-chosen and remain unfitted, and Section V-D reports a measured operating point that we deliberately did not adopt. The evaluation set has content and resolution confounds that we characterise rather than remove, so the absolute figures below are an upper bound on field performance, not an estimate of it.

II. RELATED WORK

Learned detectors. Wang et al. [2] showed that a classifier trained on the output of a single CNN generator transfers surprisingly well across architectures, establishing the single-classifier baseline that most subsequent work follows. Gragnaniello et al. [5] re-examined those claims under realistic post-processing and found the reported robustness substantially overstated once compression and resizing are applied.

For diffusion output specifically, Corvi et al. [4] showed that detectors trained on GAN imagery degrade sharply, and Ojha et al. [3] argued for features not tied to one generator family. Our classifier signal is an off-the-shelf instance of this line of work, and Section V-A shows why the specific instance matters more than the surrounding architecture.

Frequency-domain cues. Up-sampling layers in generative architectures leave periodic spectral artefacts, which Durall et al. [6] and Frank et al. [7] exploited directly. These cues are strong on the generators they were characterised on and known to be fragile under compression and rescaling. We include a deliberately simple FFT-based statistic as an independent, fully explainable signal, and Section VII reports the evidence that it is not yet measuring synthesis.

Provenance. The C2PA specification [11] defines cryptographically signed content credentials that travel inside the file. Provenance is qualitatively different from the other two signals: it is evidence about the file’s history rather than an inference from its pixels, and it is high-precision and low-recall by construction. The literature treats detection and provenance largely separately; combining them raises the availability problem that motivates this paper.

Evaluation practice. Our protocol follows the concern raised in [5] that detector evaluations frequently measure dataset artefacts. Community Forensics [8] supplies per-image generator attribution, which makes per-generator failure analysis possible rather than merely aggregate.

III. SYSTEM DESIGN

Fig. 1 gives the pipeline. The system exposes an HTTP API; a single endpoint accepts an image or a video and returns a verdict, a fused confidence, and the full per-signal breakdown that produced it.

A. The three signals

Classifier ($w = 0.60$). A pretrained image-classification head is applied to the image, or to every sampled frame of a video, and its label distribution is collapsed into $P(\text{machine-generated})$. Label vocabularies are mapped explicitly; a head reporting only generic labels such as LABEL_0/LABEL_1 is *not* mapped by guessing an index, because a wrong guess silently inverts the signal. An unrecognised vocabulary makes the signal report unavailable. The model used in all experiments below is a SwinV2 [10] head (haywoodslan/ai-image-detector-deploy, 744 MB) selected by measurement in Section V-A. It is loaded once at application start-up and never on the request path.

Provenance ($w = 0.25$). The file is inspected for two opposite kinds of evidence. A generator fingerprint—a known generator name in decoded EXIF or XMP text, or a C2PA/JUMBF manifest marker—yields score 1.0. Genuine camera EXIF (Make and Model present, no generator string) yields 0.0. A file carrying neither makes the signal report *unavailable*, which is the case that Section III-B exists to handle. Manifest detection is a bounded raw-byte scan for markers rather than a full C2PA parse, so the system can report that a manifest is

present but not who signed it; this is a stated implementation gap.

Frequency ($w = 0.15$). The image is converted to greyscale, resized to 512×512 , and its DC component removed. The 2-D FFT is taken, and the fraction of total spectral energy lying outside a central low-frequency disc of radius $0.03125 \times 512 = 16$ px is computed. That ratio is mapped linearly onto $[0, 1]$ over the interval $[0.05, 0.45]$. **Both endpoints are unfitted:** they bracket the range over which the ratio moves at all, and encode no measured boundary between generated and real. Section VII reports the calibration measurement that keeps them labelled that way.

The three signals were chosen to be independent in the *evidence* they consult—a neural judgement about pixels, a document-metadata lookup, and a spectral statistic—so that they can disagree, and so that a failure mode of one is not automatically a failure mode of another.

B. Availability-aware fusion

Let $S = \{\text{model}, \text{metadata}, \text{frequency}\}$ with fixed weights w_k , and let $A \subseteq S$ be the subset of signals that successfully produced a score for this file. The fused confidence is

$$c = \frac{\sum_{k \in A} s_k w_k}{\sum_{k \in A} w_k}, \quad c = 0.5 \quad \text{if } A = \emptyset. \quad (1)$$

Equation (1) is a weighted average over the available signals with the weights renormalised, not over all signals with missing ones zeroed. The distinction is the central design decision of the system. A score of 0.0 means “this looks real”; unavailability means “no evidence either way”. Conflating them makes every file whose metadata was stripped drift toward a verdict of authentic—and stripping metadata is exactly what every social platform does on upload. Concretely, when provenance is absent the classifier and frequency weights rescale from 0.60/0.15 to 0.80/0.20 and decide between themselves.

The $A = \emptyset$ case maps to 0.5 rather than to 0: total signal failure is maximum ignorance, and 0.5 falls inside the abstention band below, so a fully degraded system declines to answer instead of reporting “real”.

The verdict is a three-way thresholding of c : $c \geq 0.65 \Rightarrow \text{AI_GENERATED}$; $c \leq 0.35 \Rightarrow \text{LIKELY_REAL}$; otherwise UNCERTAIN. UNCERTAIN is a first-class outcome. It is the system declining to answer, and throughout this paper it is counted separately from correct and incorrect rather than forced into a confusion matrix.

C. Degradation behaviour

A failure of the classifier—the heaviest signal—is not an error response. The signal reports unavailable, (1) renormalises, and the other two still produce a verdict. This follows from the same principle as the fusion rule: a partial answer with its provenance visible is more useful than no answer. The classifier’s health is exposed on a separate status endpoint rather than by failing analysis requests.

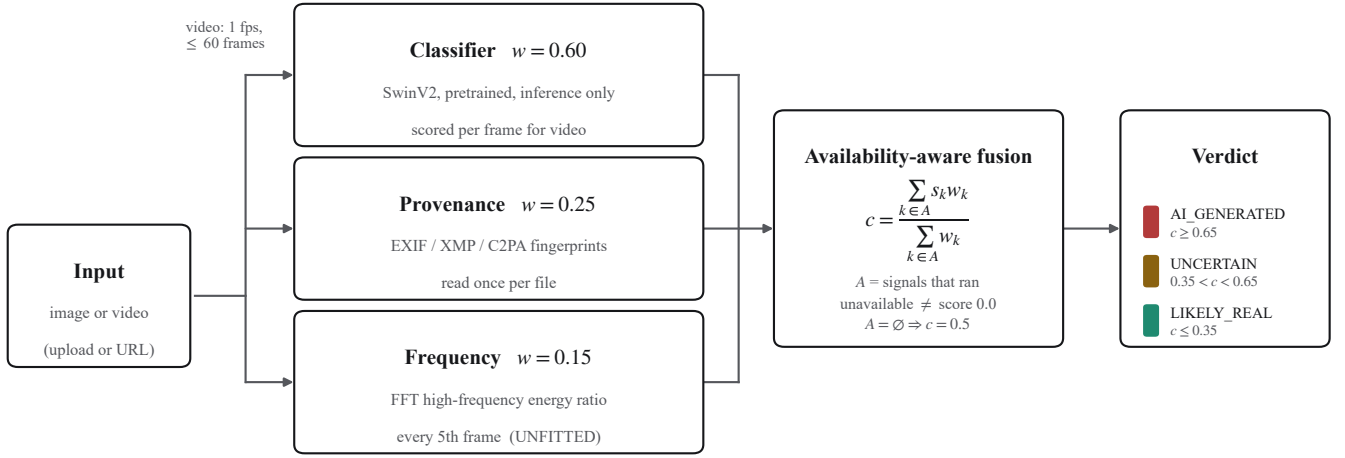


Fig. 1. Pipeline. Three independent signals are computed, then fused over only those that actually ran. Video is treated as images over time: frames are sampled at 1 fps up to a 60-frame cap, the classifier scores every sampled frame, provenance is read once from the container, and the frequency signal runs on every fifth frame.

IV. EVALUATION METHODOLOGY

A. Two tracks, never averaged

Track A (public). 400 images drawn from Community Forensics [8] (CC BY-NC-SA 4.0): 200 generated and 200 real. The generated half spans 50 distinct generators at 4 images each, all of the LatDiff architecture family; the real half is FFHQ [9].

Track B (hand-collected). 16 files, 13 labelled (4 generated, 9 real), collected by the authors: camera originals with intact EXIF, images exported from commercial generators with Content Credentials, and files deliberately routed through a messaging platform so that their metadata is stripped in transit.

The tracks are never averaged, for a structural reason. Public datasets are redistributed re-encoded: **0 of the 400 Track A images carry any EXIF**, so the provenance signal cannot fire on a single one of them. Averaging would let 400 images with a structurally dead signal drown 13 that exercise it, and the resulting “ensemble” headline would silently be a two-signal result. Track A measures the classifier and frequency signals at scale; Track B is the only place provenance does anything at all.

Track B carries a second disclosure. The 12 labelled images in it were the set used to select the classifier (Section V-A). It is therefore the model-selection set as well as the demonstration set, and its results are reported as counts and as a functional check on the provenance branches—never as an accuracy estimate.

B. Metrics and the treatment of abstention

Accuracy, precision, recall and F_1 are computed over *decided* files only, with abstentions excluded and reported separately as coverage (decided/labelled). A detector that abstains on half the set and is right on the rest is not accurate, and coverage is the honest companion to any accuracy figure. ROC-AUC is computed over all labelled files including abstentions,

since AUC ranks a continuous score and never consults a threshold; it is obtained through the rank (Mann–Whitney U) identity with ties averaged.

C. Threshold selection protocol

Thresholds are chosen on one half of Track A and reported on the other, split stratified by class and seeded. A threshold picked on the same rows it is then scored against is fitted to those rows and the resulting number estimates nothing. This was not a formality: in an earlier configuration, a search over all 400 rows at once suggested a cut near 0.06 lifting accuracy to roughly 0.73, while under a proper split the selection half chose 0.806 and reached only 0.6850 held out.

D. Resolution normalisation

Every image in Track A is analysed twice: as distributed (*naive*), and after a control we call *resolution normalisation*.

The confound is this. Real images in this set are natively 1024×1024 and generated images natively 512×512 , with no overlap. Every detector in the pipeline resizes to 512 internally, so under the naive protocol the real images arrive having been downsampled—which destroys high-frequency content—while the generated images arrive untouched. A frequency statistic can then separate the classes by measuring who was downsampled rather than who was synthesised.

Resizing everything to a common target does *not* fix this. A 1024^2 image resized to 512 is still downsampled $2\times$ while a native 512^2 image is not; the operation is identical but its effect depends on the input size, which is exactly the variable that differs by class. We instead take a **centre crop of 512×512 native pixels**. Nothing is interpolated, no frequency content is created or destroyed, and every image reaches the detectors as an unresampled 512×512 block, so the processing history is genuinely identical across classes. Crops

TABLE I

CANDIDATE CLASSIFIERS ON THE 12 LABELLED HAND-COLLECTED IMAGES (4 GENERATED, 8 REAL). INFERENCE ONLY; NO FINE-TUNING.

Model (HuggingFace identifier)	Correct	Mean, gen.	Mean, real
haywoodslloan/ai-image-detector-deploy	12/12	0.987	0.002
Ateeqq/ai-vs-human-image-detector	11/12	0.752	0.001
dima806/ai_vs_human_generated_image_detection	8/12	0.012	0.011
Organika/sdxl-detector	5/12	0.438	0.574
dima806/ai_vs_real_image_detection	4/12	0.766	0.694

are written losslessly as PNG so the control itself introduces no compression artefacts.

What the crop does not fix: it shows a sub-region of a large image and the whole of a small one, so framing still differs. That is a content difference, not a resampling one, and it is reported as such. Normalisation is refused on Track B, because re-saving as PNG discards the EXIF that Track B exists to test.

V. RESULTS

A. Classifier selection is a measurement, not a model-card decision

The classifier was initially an SDXL-specific detector chosen from its model card. Measured on the 12 labelled hand-collected images it was not merely weak but *anti-correlated*: mean score 0.438 on generated images against 0.574 on real photographs. It scored genuine iPhone originals at 0.998 “artificial” and a commercially generated image at 0.003. No weight and no threshold can repair a signal that points the wrong way.

Five cached candidates were then scored over the identical files, inference only (Table I). The separation between the best and worst is far larger than any fusion parameter could compensate for, which is the practical lesson: for a fused detector, the choice of the heaviest component dominates the choice of how to combine it.

B. Track A headline

Table II gives the aggregate results in both conditions and Fig. 2 the corresponding ROC curves.

Recall is identical in both conditions, to four decimal places, and that is not a coincidence. The generated images are natively 512², so the centre crop is a no-op on them: their per-image classifier and frequency scores are byte-identical across the two runs. Only the real images change. The entire normalised gain is therefore 31 fewer false positives on real photographs (70→39), which is exactly what removing a resampling confound should do if that confound was inflating the real side. Fig. 3 shows this directly: the two generated score distributions coincide exactly while the real distribution shifts toward zero.

TABLE II

TRACK A, $n = 400$. METRICS OVER DECIDED FILES; COVERAGE IS DECIDED/LABELLED. AUC OVER ALL LABELLED FILES.

	naive	normalised
Accuracy	0.7595	0.8384
Precision	0.7107	0.8152
Recall	0.8731	0.8731
F_1	0.7836	0.8431
ROC-AUC	0.9196	0.9487
Coverage	0.9875	0.9900
Abstentions	5	4
True positives	172	172
False positives	70	39
True negatives	128	160
False negatives	25	25

TABLE III

ABLATION ON TRACK A, RECOMPUTED FROM THE PER-SIGNAL DUMP. ROWS MARKED † ARE ARITHMETIC IDENTITIES FORCED BY THE DATASET, NOT INDEPENDENT RESULTS.

Configuration	naive		normalised	
	Acc.	AUC	Acc.	AUC
Full ensemble	0.7595	0.9196	0.8384	0.9487
Classifier only	0.7563	0.8397	0.8367	0.8947
Provenance only	—	0.5000	—	0.5000
Frequency only	0.7194	0.9012	0.7177	0.9129
— Provenance†	0.7595	0.9196	0.8384	0.9487
— Frequency†	0.7563	0.8397	0.8367	0.8947

The system’s characteristic error is calling a real FFHQ portrait generated, not letting a generated image pass: precision (0.8152) trails recall (0.8731) in the normalised condition.

C. Ablation

Table III re-fuses the dumped per-signal scores under six configurations. No inference is re-run: dropping a signal from a configuration is arithmetically the same operation as that signal being unavailable at runtime, so the ablation exercises the real fusion path rather than a parallel reimplement of it.

Three points must be read off this table carefully.

The six rows contain only three distinct results. Because provenance is available on 0 of 400 files, “ensemble — provenance” is arithmetically identical to the full ensemble, and “ensemble — frequency” is identical to classifier-only. These are identities forced by the dataset and must not be presented as independent findings. They are also the clearest possible argument for why Track B exists.

The provenance row is not a measurement. With no signal available every image abstains, so accuracy, precision, recall and F_1 are undefined, and the AUC of 0.5000 is the definition of a constant predictor.

The frequency signal is extremely lopsided: precision 1.0000 at recall 0.3237 in both conditions. It never produces a false positive on this set and catches roughly a third of

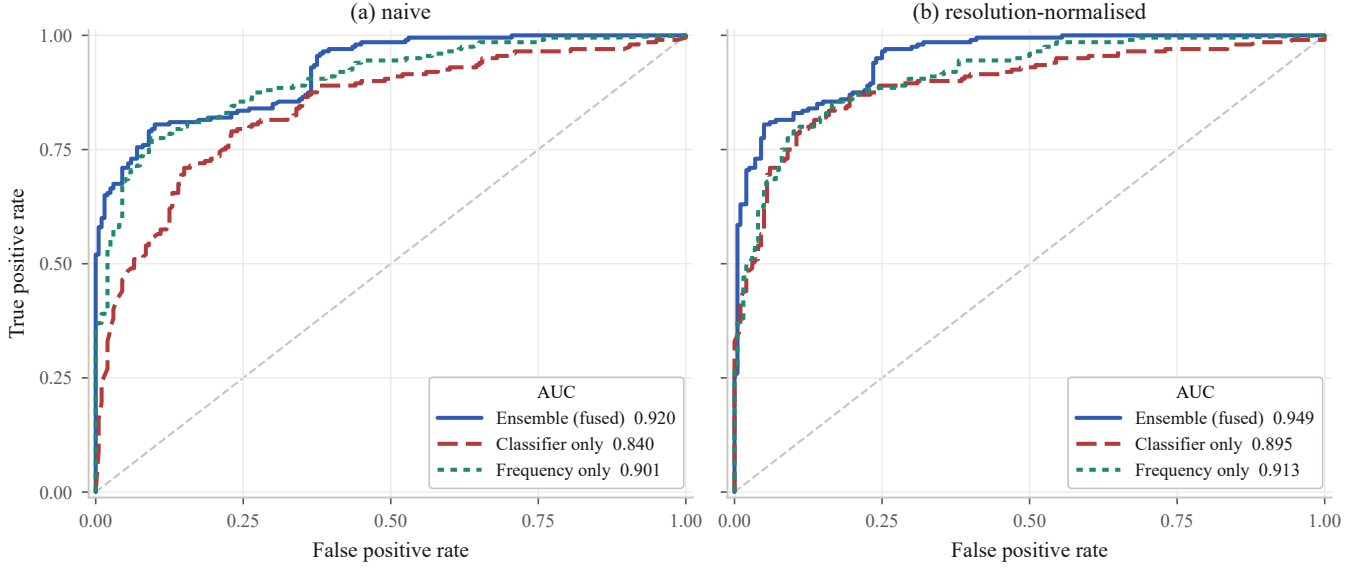


Fig. 2. ROC curves on Track A, $n = 400$, for the fused confidence and for two of its constituents taken alone. The provenance signal is omitted: it is unavailable on all 400 files and is a constant predictor here. The fused ensemble dominates both constituents in both conditions.

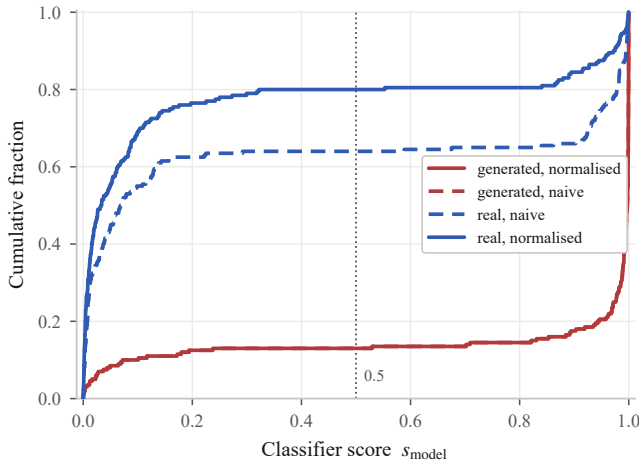


Fig. 3. Empirical CDF of the classifier score by class and condition, Track A. The two generated curves coincide exactly—the centre crop does not touch a natively 512×512 image—while the real distribution moves toward zero under normalisation. The fraction of real images scoring above 0.5 falls from 72/200 to 40/200.

generated images—the profile of a signal that fires confidently on a visually distinct subset rather than one that generalises.

D. Threshold selection

Table IV reports operating points chosen on the selection half and scored on the held-out half.

Both measured operating points survive the selection/reporting split, and neither was adopted. The band improves held-out accuracy from 0.8342 to 0.9664 but costs 25 points of coverage—abstaining on a quarter of inputs instead of one in two hundred—which is a different product, not a

TABLE IV
THRESHOLD SELECTION, TRACK A. CHOSEN ON ONE STRATIFIED HALF ($n = 200$), REPORTED ON THE OTHER ($n = 200$). NEITHER MEASURED OPERATING POINT WAS ADOPTED.

Cond.	Operating point	Sel. acc.	Held-out	Cov.
naive	single cut 0.808	0.8350	0.8650	1.000
naive	band 0.14/0.82	0.9220	0.9444	0.720
naive	<i>current</i> 0.35/0.65	—	0.7437	0.995
norm.	single cut 0.808	0.8700	0.8800	1.000
norm.	band 0.10/0.82	0.9586	0.9664	0.745
norm.	<i>current</i> 0.35/0.65	—	0.8342	0.995

tuned one. The single cut is the more uncomfortable result: it improves held-out accuracy at *no* coverage cost, and says plainly that the hand-chosen 0.65 threshold sits too low for this classifier’s score distribution, because the real portraits it is unsure about fuse into the 0.65–0.80 range and are called generated.

We declined both because they are fitted to a score distribution shaped by the dataset confounds of Section VII—face photographs versus digital art—which is not what the intended inputs look like, and because on the 13 hand-collected files the current band makes no error. There is therefore evidence that the threshold is wrong *on FFHQ*, and no field evidence that it is wrong in general. Adopting it would trade a measurable gain on a confounded set for unknown behaviour on real inputs. The finding is recorded; the constants are unchanged.

E. Per-generator failure analysis

Of 50 generators, none was missed on every one of its four images, 29 were caught on all four, and the worst slip rate is 2 of 4 on seven generators whose mean classifier scores lie

between 0.457 and 0.646—near the decision boundary rather than at zero. Nine of the ten worst misclassifications by margin are generated images called real; the tenth is an FFHQ portrait scored 0.998 by the classifier and fused to 0.889.

This error structure differs qualitatively from that of the original classifier, under which six generators were never caught once and eight always were—a split with a clear mechanism, namely an SDXL-specific detector recognising SDXL-family output and passing SD 1.5-era fine-tunes and textual inversions at near-zero confidence. The replacement misses thinly and everywhere instead, which is the signature of a general-purpose detector operating near its margin rather than of a specialist outside its training family.

VI. TWO NEGATIVE RESULTS

Both results in this section were invisible to unit tests and to aggregate metrics. Both were surfaced by the same mechanism: an evaluation harness that records every signal’s score *and* its availability flag for every file, rather than only the fused verdict.

A. A three-character string inverted the provenance signal

The generator signature list contained the string "veo"—three characters. In a few hundred kilobytes of compressed image data, a three-character case-insensitive byte sequence occurs by chance with high probability. The provenance signal consequently reported “generator signature found” at score 1.0—full confidence—on **62 of 200 real photographs** (31.0%), and across the set fired on 65 real images against 23 generated ones. The signal intended to prove synthetic origin was firing more often on real photographs than on generated ones, and contributing a quarter of the fused weight toward AI_GENERATED every time it did.

Three aspects are worth drawing out. First, every unit-level behaviour was correct: the scan found the string it was asked to find, the API returned well-formed responses, and the detail string was plausible. Second, the fix is a separation of concerns rather than a deletion—signature matching now uses the full list for decoded EXIF and XMP text, where a short generator name is a word, and a strict list requiring six or more characters plus structural C2PA box markers for raw binary. Verified afterwards on the same 200 real images: 0 false positives. Third, the fix has a stated cost: a video whose only evidence is the bare string “sora” in its container is no longer detected. That trade is deliberate—a signal that fires on a third of all real photographs is worse than one with a known blind spot.

B. The ensemble once ranked below its lowest-weighted component

With the original classifier, the fused ensemble reached AUC 0.7918 naive and 0.8461 normalised while frequency alone—carrying the smallest weight, 0.15—reached 0.9012 and 0.9129. The fused system was worse than its cheapest constituent. We reported that result rather than reweighting,

because reweighting toward frequency on this dataset would have been fitting to the content confound of Section VII.

After the classifier replacement of Section V-A, the ordering reverses: the ensemble reaches 0.9196 and 0.9487 and dominates every constituent (Table III). The instructive part is what did *not* change. The frequency signal is the same code on the same files and reports the same AUC to four decimal places in both eras. The fusion rule and the weights are unchanged. Only the classifier moved.

The conclusion we draw is deliberately modest. Unfitted weights did not become correct; a classifier that ranked poorly on its own dragged a 0.60-weighted average below the frequency signal, and one that ranks well combines with it to something better than either. An ensemble is not robust to a bad heaviest component, and an ablation table is what makes that visible.

VII. LIMITATIONS

Dataset confounds. Three properties of Track A inflate its numbers. (i) Zero EXIF across all 400 images, so provenance is structurally dead. (ii) Every real image is an FFHQ face—aligned, centred portraits—while the generated half is varied digital art, illustration and photography, so a detector can separate the classes partly on subject matter. (iii) The resolution asymmetry of Section IV-D, which the normalised condition controls. Together these make Track A’s figures an upper bound.

The frequency signal is not yet measuring synthesis. Its endpoints are unfitted, and a calibration run over the hand-collected files gives raw ratios of 0.0394–0.3251 with the real files spanning 0.0499–0.2357 and the generated files 0.1101–0.3251: substantially overlapping ranges. Its discrimination on Track A is not explained by resampling—under normalisation it rose rather than fell, 0.9012 to 0.9129—but it cannot yet be attributed to synthesis artefacts rather than to subject matter, and the centre crop plausibly intensified the content confound, since a 512² crop of an FFHQ portrait is mostly smooth skin while the generated images are detailed scenes. The ensemble’s margin over the classifier alone (+0.080 naive, +0.054 normalised) arrives through this signal and inherits the same caveat. If a content-matched set shows the signal does not separate the classes, setting its weight to zero and reporting that is the correct outcome.

Weights remain unfitted. 0.60/0.25/0.15 were chosen by inspection before any measurement existed. That they now produce a better result than any single signal is evidence about the classifier, not about the weights.

Track B is small and doubly used. 13 labelled files demonstrate that each provenance branch works; they do not measure how often it works. Since those files also selected the classifier, their perfect result is not an estimate.

Implementation gaps. Real C2PA manifest parsing is not implemented, so the system reports that a manifest is present but never who signed it or whether the claim chain validates. Video container metadata uses the same bounded raw scan

rather than MP4 atom parsing. Clips longer than 60 seconds are analysed over their first 60 seconds only.

Cost. The classifier is a 744 MB SwinV2 on CPU. In-process, a 512×512 image takes 1.29 s on average (4-core i5-1135G7, threads capped at half). End-to-end through the API, a full-resolution camera original takes 4–6 s, a 6-frame clip 18.8 s, and a 13-frame clip 46.6 s; a 60-frame clip extrapolates to 3.5–4 minutes and has not been timed. This is roughly an order of magnitude slower per image than the model it replaced, and it makes video analysis interactive only for short clips.

No video is evaluated at scale. Track A is images only; the frame-sampling and aggregation path is exercised by a single clip in Track B and by plumbing fixtures.

VIII. CONCLUSION AND FUTURE WORK

We presented a three-signal detector for machine-generated images and video whose fusion rule treats an unavailable signal as absent evidence rather than as evidence of authenticity, and evaluated it under a protocol designed to keep dataset artefacts separable from detection. The fused system reaches ROC-AUC 0.9196 on a 400-image public set spanning 50 generators and 0.9487 under a resolution control, dominating each constituent signal; the entire gain from the control is a reduction in false positives on real photographs, with recall unchanged.

The methodological claims are the ones we would most like carried forward. Per-signal score dumping is what exposed a provenance signature that fired at full confidence on 31% of real photographs while every unit test passed. An ablation table is what exposed a fused ensemble ranking below its lowest-weighted component. Neither is visible in an aggregate accuracy figure, and both are cheap to produce. We would argue they belong in the minimum reporting standard for any fused detector.

Future work, in priority order: build a content- and resolution-matched evaluation set, which is the prerequisite for attributing the frequency signal’s discrimination and therefore for fitting the weights at all; fit the three weights on that set and revisit the operating points of Section V-D; implement real C2PA manifest parsing so the system can report a signing authority rather than mere presence; assemble a labelled video set, since the temporal path is currently almost unmeasured; and collect a fresh hand-collected set not used to select any component, so that a field-realistic estimate becomes possible.

REPRODUCIBILITY

Every figure and table in this paper is produced by a script in the project repository, and no number was computed by hand. The evaluation harness writes a per-file, per-signal dump (score and availability flag for each of the three signals) from which the ablation and threshold-selection tables are recomputed without re-running inference. The evaluation-set builder records the generator and architecture behind every generated image, which is what makes the per-generator failure analysis possible.

One reporting detail, for exactness. The metrics harness ranks the fused confidence as the API emits it, rounded to four decimal places, which leaves ties; the ablation recomputes from the six-decimal per-signal dump and breaks them. The two differ by 2.5×10^{-5} in the naive condition and agree exactly in the normalised one. All AUC figures in this paper are quoted from the ablation recomputation so that a single source is used throughout.

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